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Systems and methods for processing objects including an auto-shuttle system

Abstract

A storage, retrieval and processing system for processing objects is disclosed. The storage, retrieval and processing system includes a plurality of storage bins providing storage of a plurality of objects, where the plurality of storage bins is in communication with a retrieval conveyance system, a programmable motion device in communication with the retrieval conveyance system for receiving the storage bins from the plurality of bins, where the programmable motion device includes an end effector for grasping and moving a selected object out of a selected storage bin, and a movable carriage for receiving the selected object from the end effector of the programmable motion device, and for carrying the selected object to one of a plurality of destination bins.

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Background/Summary

PRIORITY (1) The present application is a continuation of U.S. patent application Ser. No. 17/739,735, filed May 9, 2022; which is a continuation of U.S. patent application Ser. No. 16/744,862, filed Jan. 16, 2020, now U.S. Pat. No. 11,365,051, issued Jun. 21, 2022; which is a continuation of U.S. patent application Ser. No. 15/924,883, filed Mar. 19, 2018, now U.S. Pat. No. 10,583,553, issued Mar. 10, 2020, which claims priority to U.S. Provisional Patent Application Ser. No. 62/473,843, filed Mar. 20, 2017, the disclosures of which are hereby incorporated by reference their entireties.

BACKGROUND

- (1) The invention generally relates to storage and retrieval systems, and relates in particular to automated storage and retrieval systems that are used with systems for processing objects.
- (2) Automated storage and retrieval systems (AS/RS) generally include computer controlled systems of automatically storing (placing) and retrieving items from defined storage locations. Traditional AS/RS typically employ totes (or bins), which are the smallest unit of load for the system. In these systems, the totes are brought to people who pick individual items out of the totes. When a person has picked the required number of items out of the tote, the tote is then re-inducted back into the AS/RS.
- (3) In these traditional systems, the totes are brought to a person, and the person may either remove an item from the tote or add an item to the tote. The tote is then returned to the storage location. Such systems, for example, may be used in libraries and warehouse storage facilities. The AS/RS involves no processing of the items in the tote, as a person processes the objects when the tote is brought to the person. This separation of jobs allows any automated transport system to do what it is good at—moving totes—and the person to do what the person is better at—picking items out of cluttered totes. It also means the person may stand in one place while the transport system brings the person totes, which increases the rate at which the person can pick goods.
- (4) There are limits however, on such conventional systems in terms of the time and resources required to move totes toward and then away from each person, as well as how quickly a person can process totes in this fashion in applications where each person may be required to process a large number of totes. There remains a need therefore, for an AS/RS that stores and retrieves objects more efficiently and cost effectively, yet also assists in the processing of a wide variety of objects.

SUMMARY

- (5) In accordance with an embodiment, the invention provides a storage, retrieval and processing system for processing objects. The storage, retrieval and processing system includes a plurality of

storage bins providing storage of a plurality of objects, where the plurality of storage bins is in communication with a retrieval conveyance system, a programmable motion device in communication with the retrieval conveyance system for receiving the storage bins from the plurality of bins, where the programmable motion device includes an end effector for grasping and moving a selected object out of a selected storage bin, and a movable carriage for receiving the selected object from the end effector of the programmable motion device, and for carrying the selected object to one of a plurality of destination bins.

(6) In accordance with another embodiment, the invention provides a storage, retrieval and processing system for processing objects, where the storage, retrieval and processing system includes a plurality of storage bins, at least one programmable motion device, and a movable carriage. The plurality of storage bins provides storage of a plurality of objects, and is provided in at least two linear arrangements, each of which is in communication with a retrieval conveyance system. The at least one programmable motion device is in communication with the retrieval conveyance system for receiving the storage bins from the plurality of bins, and the programmable motion device is for grasping and moving a selected object out of a selected storage bin. The reciprocating movable carriage is for receiving the selected object from the end effector of the programmable motion device, and for carrying the selected object to one of a plurality of destination bins that are provided in at least two linear arrangements of destination bins.

(7) In accordance with a further embodiment, the invention provides a method of providing storage, retrieval and processing of objects. This includes the steps of providing a plurality of storage bins for storing a plurality of objects, the plurality of storage bins being in communication with a retrieval conveyance system, receiving the storage bins from the plurality of storage bins at a programmable motion device in communication with the retrieval conveyance system, grasping and moving a selected object out of a selected storage bin; receiving the selected object from the programmable motion device and carrying the selected object to one of a plurality of destination bins.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The following description may be further understood with reference to the accompanying drawing in which:
- (2) FIG. 1 shows an illustrative diagrammatic view of a storage, retrieval and processing system in accordance with an embodiment of the present invention;
- (3) FIG. 2 shows an illustrative diagrammatic view of a storage section portion of the storage, retrieval and processing system of FIG. 1;
- (4) FIGS. 3A and 3B show illustrative diagrammatic views of embodiments of a bin displacement system for use in a retrieval conveyance system of the invention;
- (5) FIG. 4 shows an illustrative diagrammatic exploded view of a box assembly for use as a storage bin or destination bin in accordance with various embodiments of the present invention;
- (6) FIG. 5 shows the assembled box tray assembly of FIG. 4;
- (7) FIG. 6A-6D show illustrative diagrammatic views of a further embodiment of a bin displacement system for use in further embodiments of the invention;
- (8) FIG. 7 shows an illustrative diagrammatic view of a programmable motion device processing station for use in a storage, retrieval and processing system in accordance with an embodiment of the present invention;
- (9) FIG. 8 shows an illustrative diagrammatic view from the perception system of FIG. 7, showing a view of objects within a bin of objects to be processed;
- (10) FIGS. 9A and 9B show illustrative diagrammatic views of a grasp selection process in a

- storage, retrieval and processing system of the embodiment of the present invention;
- (11) FIGS. **10A** and **10B** show illustrative diagrammatic views of a grasp planning process in a storage, retrieval and processing system of the embodiment of the present invention;
- (12) FIGS. **11A** and **11B** show illustrative diagrammatic views of a grasp execution process in a storage, retrieval and processing system of the embodiment of the present invention;
- (13) FIG. **12** shows an illustrative diagrammatic view of a portion of the processing system together with a portion of a destination section of an embodiment of a storage, retrieval and processing system in accordance with an embodiment of the invention;
- (14) FIG. **13** shows an illustrative diagrammatic view of a processing section in a storage, retrieval and processing system in accordance with an embodiment of the invention wherein an object is placed in a carriage;
- (15) FIG. **14** shows an illustrative diagrammatic view of the processing section of FIG. **13** with the carriage having been moved along its track;
- (16) FIG. **15** shows an illustrative diagrammatic view of the processing section of FIG. **13** with the carriage having transferred its load to a destination bin;
- (17) FIGS. **16A** and **16B** show illustrative diagrammatic views of a bin removal mechanism for use in a storage, retrieval and processing system in accordance with an embodiment of the invention;
- (18) FIG. **17** shows an illustrative diagrammatic view of the processing section of FIG. **13** with the carriage having returned to its base, and a removed destination bin being moved urged from its location;
- (19) FIG. **18** shows an illustrative diagrammatic view of the processing section of FIG. **13** with the removed destination bin being moved along an output conveyor;
- (20) FIG. **19** shows an illustrative diagrammatic view of a storage, retrieval and processing system in accordance with a further embodiment of the present invention employing a plurality of storage sections, processing sections, and a plurality of destination sections; and
- (21) FIG. **20** shows an illustrative diagrammatic view of a storage, retrieval and processing system in accordance with a further embodiment of the present invention employing further pluralities of storage sections, processing sections and destination sections.
- (22) The drawings are shown for illustrative purposes only.

DETAILED DESCRIPTION

- (23) In accordance with an embodiment, the invention provides a storage, retrieval and processing system for processing objects. The system includes a plurality of storage bins providing storage of a plurality of objects, a programmable motion device, and a movable carriage. The plurality of storage bins is in communication with a retrieval conveyance system. The programmable motion device is in communication with the retrieval conveyance system for receiving the storage bins from the plurality of bins, and the programmable motion device includes an end effector for grasping and moving a selected object out of a selected storage bin. The movable carriage receives the selected object from the end effector of the programmable motion device, and carries the selected object to one of a plurality of destination bins.
- (24) With reference to FIG. **1**, a system **10** in accordance with an embodiment of the present invention includes a storage section **12** at which a plurality of storage bins **14** are provided, a processing section **16** that includes a programmable motion device **18**, and one or more destination section **20** that include one or more shuttle carriers **22**. Generally, storage bins **14** are selectively provided to and from the processing section **16**, where objects are moved from a selected storage bin and placed or dropped by the programmable motion device **18** into a carriage **22** of the destination section **20** for delivery to one of a plurality of destination bins **24**. When a destination bin **24** is complete, the completed bin is urged onto an output conveyor **26** for further processing or transport.
- (25) As shown in FIG. **2**, the storage bins **14** of the storage section **12** are provided adjacent to retrieval conveyors **28** that carry a selected storage bin toward the processing station **16**. As further

shown with reference to FIGS. 3A and 3B, when a particular storage bin **15** is selected for retrieval, a bin removal mechanism **30** travels along between the bins **14**, and stops adjacent to the selected storage bin as shown in FIG. 3A.

(26) With reference to FIG. 3B, the system will then move an urging member **32** of the mechanism **30** to push the selected bin **15** onto a conveyor **28** leading to the processing section **16**. The removal mechanism **30** may actuate the urging member **32** by any of a variety of processes, including having the support beam **31** be threaded with the urging member **28** being threaded onto the beam **29** such that it moves when the support beam is rotated, or by other mechanical, pneumatic or electronic actuation.

(27) The conveyor **28** (as well as the other conveyors in the system) may be motion controlled so that both the speed and the direction of the conveyor (e.g., rollers or belt) may be controlled. In certain embodiments, certain of the conveyors (e.g., leading from the storage station **12** to the processing station **16**) may be gravity biased to cause any storage bin on any conveyor system to be delivered to the processing section **16** near the programmable motion device **18**.

(28) The bins **14** may be provided as boxes, totes, containers or any other type of device that may receive and hold an item. In further embodiments, the bins may be provided in uniform trays (to provide consistency of spacing and processing) and may further include open covers that may maintain the bin in an open position, and may further provide consistency in processing through any of spacing, alignment, or labeling.

(29) For example, FIG. 4 shows an exploded view of a box tray assembly **130**. As shown, the box **132** (e.g., a standard shipping sized cardboard box) may include bottom **131** and side edges **133** that are received by a top surface **135** and inner sides **137** of a box tray **134**. The box tray **134** may include a recessed (protected) area in which a label or other identifying indicia **146** may be provided, as well as a wide and smooth contact surface **151** that may be engaged by an urging or removal mechanism as discussed below.

(30) As also shown in FIG. 4, the box **132** may include top flaps **138** that, when opened as shown, are held open by inner surfaces **140** of the box cover **136**. The box cover **136** may also include a recessed (protected) area in which a label or other identifying indicia **145** may be provided. The box cover **136** also provides a defined rim opening **142**, as well as corner elements **144** that may assist in providing structural integrity of the assembly, and may assist in stacking un-used covers on one another. Un-used box trays may also be stacked on each other.

(31) The box **132** is thus maintained securely within the box tray **134**, and the box cover **136** provides that the flaps **138** remain down along the outside of the box permitting the interior of the box to be accessible through the opening **142** in the box cover **136**. FIG. 5 shows a width side view of the box tray assembly **130** with the box **132** securely seated within the box tray **134**, and the box cover holding open the flaps **138** of the box **132**. The box tray assemblies may be used as any or both of the storage bins and destination bins in various embodiments of the present invention.

(32) With reference to FIGS. 6A-6D, a box kicker **184** in accordance with an embodiment of the present invention may be suspended by and travel along a track **86**, and may include a rotatable arm **88** and a roller wheel **190** at the end of the arm **88**. With reference to FIGS. 6B-6D, when the roller wheel **190** contacts the kicker plate **151** (shown in FIG. 4) of a box tray assembly **120**, the arm **188** continues to rotate, urging the box tray assembly **180** from a first conveyor **182** to a second conveyor **180**. Again, the roller wheel **190** is designed to contact the kicker plate **151** of a box tray assembly **181** to push the box tray assembly **181** onto the conveyor **180**. Such a system may be used to provide that boxes that are empty or finished being unloaded may be removed (e.g., from conveyor **182**), or that boxes that are full or finished being loaded may be removed (e.g., from conveyor **182**). The conveyors **180**, **182** may also be coplanar, and the system may further include transition roller **183** to facilitate movement of the box tray assembly **181**, e.g., by being activated to pull the box tray over to the conveyor **180**.

(33) With reference to FIG. 7, the processing station **16** also includes a perception unit **36** that is

mounted adjacent to the base of the programmable motion device **18**, and looks down into the selected storage bin. An end effector **38** of the programmable motion device grasps an object in bin, and moves to deliver the object to a desired destination bin **24** by placing or dropping the object into a carriage, that then shuttles the object to the selected destination bin **24** (as shown further in FIG. **12**).

(34) The storage bin may then be returned to the plurality of storage bins at the storage station, and may be returned anywhere among the bins as long as the system knows where the bin has been returned, and knows how each of the bins may have been moved when the selected storage bin was transferred to the conveyor **28**. The storage bins, for example, may be biased (e.g., by gravity) to stack against one of the ends of each row of bins, and the returned bins may be put at the uphill end of a row of storage bins. The storage bins may be returned to the storage section **12** by the conveyors as discussed above, and may be returned to the central area between the conveyors **28** either by human personnel or by employing additional removal mechanisms as discussed above to urge bins back into the central area.

(35) FIG. **8** shows an image view **50** of a bin **14** from the perception unit **38**. The image view shows the bin **14** (e.g., on the conveyor), and the bin **14** contains objects **78**, **80**, **82**, **84** and **86**. In the present embodiment, the objects are homogenous, and are intended for distribution to different distribution packages. Superimposed on the objects **78**, **80**, **82**, **84**, **86** (for illustrative purposes) are anticipated grasp locations **79**, **81**, **83** and **85** of the objects. Note that while candidate grasp locations **79**, **83** and **85** appear to be good grasp locations, grasp location **81** does not because its associated object is at least partially underneath another object. The system may also not even try to yet identify a grasp location for the object **84** because the object **84** is too obscured by other objects. Candidate grasp locations may be indicated using a 3D model of the robot end effector placed in the location where the actual end effector would go to use as a grasp location as shown in FIG. **5**. Grasp locations may be considered good, for example, if they are close to the center of mass of the object to provide greater stability during grasp and transport, and/or if they avoid places on an object such as caps, seams etc. where a good vacuum seal might not be available.

(36) If an object cannot be fully perceived by the detection system, the perception system considers the object to be two different objects, and may propose more than one candidate grasps of such two different objects. If the system executes a grasp at either of these bad grasp locations, it will either fail to acquire the object due to a bad grasp point where a vacuum seal will not occur, or will acquire the object at a grasp location that is very far from the center of mass of the object and thereby induce a great deal of instability during any attempted transport. Each of these results is undesirable.

(37) If a bad grasp location is experienced, the system may remember that location for the associated object. By identifying good and bad grasp locations, a correlation is established between features in the 2D/3D images and the idea of good or bad grasp locations. Using this data and these correlations as input to machine learning algorithms, the system may eventually learn, for each image presented to it, where to best grasp an object, and where to avoid grasping an object.

(38) As shown in FIGS. **9A** and **9B**, the perception system may also identify portions of an object that are the most flat in the generation of good grasp location information. In particular, if an object includes a tubular end and a flat end such as object **87**, the system would identify the more flat end as shown at **88** in FIG. **9B**. Additionally, the system may select the area of an object where a UPC code appears, as such codes are often printed on a relatively flat portion of the object to facilitate scanning of the barcode.

(39) FIGS. **10A** and **10B** show that for each object **90**, **92**, the grasp selection system may determine a direction that is normal to the selected flat portion of the object **90**, **92**. As shown in FIGS. **11A** and **11B**, the robotic system will then direct the end effector **94** to approach each object **90**, **92** from the direction that is normal to the surface in order to better facilitate the generation of a good grasp on each object. By approaching each object from a direction that is substantially normal

to a surface of the object, the robotic system significantly improves the likelihood of obtaining a good grasp of the object, particularly when a vacuum end effector is employed.

(40) The invention therefore provides in certain embodiments that grasp optimization may be based on determination of surface normal, i.e., moving the end effector to be normal to the perceived surface of the object (as opposed to vertical picks), and that such grasp points may be chosen using fiducial features as grasp points, such as picking on a barcode, given that barcodes are almost always applied to a flat spot on the object.

(41) FIG. 12 shows the processing section 16 and portions of the destination sections 20 of the system, which include movable carriages 22 that may receive an object from the end effector 38 of the programmable motion device 18. The movable carriage is reciprocally movable between the destination bins 24, and as further shown in FIGS. 13 and 14, each carriage 22 moves along a track 42, and may be actuated to drop an object 44 into a desired destination bin 24 (as shown in FIG. 15).

(42) The destination bins may be provided in a conveyor (e.g., rollers or belt), and may be biased (for example by gravity) to urge all destination bins toward one end (for example, the distal end 38 as shown). With reference to FIGS. 16A and 16B, when a destination bin 24 is selected for removal (e.g., because the bin is full or otherwise ready for further processing), the system will urge the completed bin onto an output conveyor 26 to be brought to a further processing or shipment station. The conveyor 26 may be biased (e.g., by gravity) to cause any bin on the conveyor to be brought to an output location.

(43) FIGS. 16A and 16B show a bin being urged from the plurality of destination bins, onto the output conveyor 26 by the use of a displacement mechanism 48. In accordance with further embodiments, other displacement mechanisms may be used, including for example, those discussed above with reference to FIGS. 3A-3B, and 6A-6D. The destination bins may be provided as boxes or containers or any other type of device that may receive and hold an item, including the box tray assemblies discussed above with reference to FIGS. 4 and 5.

(44) Following displacement of the bin onto the conveyor (as shown in FIG. 17), each of the destination bins may be urged together, and the system will record the change in position of any of the bins that moved (as shown in FIG. 18). This way, a new empty bin may be added to the end, and the system will record the correct location and identified processing particulars of each of the destination bins. This is shown in FIGS. 13 and 14, whereby a bin 24 is removed and pushed onto the conveyor 26 (see FIG. 13), and then the remaining bins are urged together while the removed bin rolls toward an output location (see FIG. 14).

(45) Systems of the invention are highly scalable in terms of sorts-per-hour as well as the number of storage bins and destination bins that may be available. FIG. 19 shows a system 100 in accordance with a further embodiment of the present invention that includes plurality of storage sections 112, a plurality of processing sections 116, and a plurality of destination sections 120. Generally, both storage sections 112 function as discussed above with regard to FIGS. 1-6D and feed both processing sections 116 as discussed above with reference to FIGS. 1 and 4. Each processing section 116 includes a programmable motion device 118 that provides objects to carriages of one or more processing sections 120.

(46) FIG. 20 shows a system 200 in accordance with a further embodiment of the present invention that includes a further plurality (e.g., four) of storage sections 212, a further plurality (e.g., four) of processing sections 216, and a further plurality (e.g., four pairs) of destination sections 120. Generally, all storage sections 212 function as discussed above with regard to FIGS. 1-3B and feed all processing sections 216 as discussed above with reference to FIGS. 1 and 4. Each processing section 216 includes a programmable motion device 218 that provides objects to carriages of one or more processing sections 220.

(47) Control of each of the systems 10, 100 and 200 may be provided by the computer system 60 that is in communication with the storage conveyors and displacement mechanism(s), the

processing conveyors and displacement mechanism(s), and the programmable motion device(s). The computer system **60** also contains the knowledge (continuously updated) of the location and identity of each of the storage bins, and contains the knowledge (also continuously updated) of the location and identity of each of the destination bins. The system therefore, directs the movement of the storage bins and the destination bins, and retrieves objects from the storage bins, and distributes the objects to the destination bins in accordance with an overall manifest that dictates which objects must be provided in which destination boxes for shipment, for example, to distribution or retail locations.

(48) Those skilled in the art will appreciate that numerous modifications and variations may be made to the above disclosed embodiments without departing from the spirit and scope of the present invention.

Claims

1. An automated storage, retrieval and processing system comprising: an input bin storage section that includes a plurality of input bins for access by a retrieval conveyance system that brings an input bin of the plurality of input bins to a processing station; a receiving area at the processing station for receiving the input bin containing one or more objects, the processing station including a programmable motion device for grasping and removing a selected object of the one or more objects from the input bin and moving and dropping the selected object; a carrier for receiving the dropped selected object from the programmable motion device, said carrier being adapted to move the selected object as well as being adapted to discharge the selected object from the carrier; and an array of destination locations below the carrier such that the carrier is movable at an elevation above the array of destination locations, each destination location including an open top such that the carrier deposits the selected object into a selected destination location of the plurality of destination locations by discharging the selected object into the open top of the selected destination location.
2. The automated storage, retrieval and processing system as claimed in claim 1, wherein the programmable motion device includes an articulated arm with a vacuum end-effector.
3. The automated storage, retrieval and processing as claimed in claim 1, wherein the array of destination locations is provided as at least one pair of rows of destination locations.
4. The automated storage, retrieval and processing system as claimed in claim 1, wherein each destination location is adapted for sealing and shipment for transport.
5. The automated storage, retrieval and processing system as claimed in claim 1, wherein each destination location is provided adjacent an output conveyance system for receiving completed destination containers.
6. The automated storage, retrieval and processing system as claimed in claim 5, further comprising an output bin displacement system for displacing the completed destination containers onto the output conveyance system.
7. The automated storage, retrieval and processing system as claimed in claim 1, wherein the retrieval conveyance system includes an infeed conveyor section that brings any of the plurality of input bins to the processing station.
8. The automated storage, retrieval and processing system as claimed in claim 7, wherein the infeed conveyor section is gravity biased toward the processing station.
9. The automated storage, retrieval and processing system as claimed in claim 8, wherein the retrieval conveyance system includes a bin displacement system for displacing the input bin onto the infeed conveyor section.
10. The automated storage, retrieval and processing system as claimed in claim 9, wherein the retrieval conveyance system includes a return conveyor section that returns any of the input bins to the input bin storage section.

11. An automated storage, retrieval and processing system for processing objects, the automated storage, retrieval and processing system comprising: an input bin storage section that includes a plurality of input bins for access by a retrieval conveyance system that brings each input bin to the processing station; a receiving area at the processing station for receiving an input bin containing one or more objects, the processing station including a programmable motion device for grasping and removing a selected object of the one or more objects from the input bin and moving the selected object; a carrier for receiving the selected object from the programmable motion device, said carrier being adapted to move the selected object as well as being adapted to discharge the selected object from the carrier; and an array of destination locations below the carrier such that the carrier is movable at an elevation above the array of destination locations, each destination location including an open top such that the carrier deposits the selected object into a selected destination location of the plurality of destination locations by tipping the carrier to discharge the selected object into the selected destination location.
12. The automated storage, retrieval and processing system as claimed in claim 11, wherein the programmable motion device includes an articulated arm with a vacuum end-effector.
13. The automated storage, retrieval and processing system as claimed in claim 11, wherein the array of destination locations is provided as at least one pair of rows of destination locations.
14. The automated storage, retrieval and processing system as claimed in claim 11, wherein each destination location is adapted for sealing and shipment for transport.
15. The automated storage, retrieval and processing system as claimed in claim 11, wherein each destination location is provided adjacent an output conveyance system for receiving completed destination containers.
16. The automated storage, retrieval and processing system as claimed in claim 15, further comprising an output bin displacement system for displacing the completed destination containers onto the output conveyance system.
17. The automated storage, retrieval and processing system as claimed in claim 11, wherein the retrieval conveyance system includes an infeed conveyor section that brings any of the plurality of input bins to the processing station.
18. The automated storage, retrieval and processing system as claimed in claim 17, wherein the infeed conveyor section is gravity biased toward the processing station.
19. The automated storage, retrieval and processing system as claimed in claim 18, wherein the retrieval conveyance system includes a bin displacement system for displacing an input bin onto the infeed conveyor section.
20. The automated storage, retrieval and processing system as claimed in claim 19, wherein the retrieval conveyance system includes a return conveyor section that returns any of the input bins to the input bin storage section.
21. A method of storing, retrieving and processing objects, said method comprising: storing a plurality of input bins at an input bin storage section; moving a selected input bin of the plurality of input bins to a retrieval conveyance system; receiving, at a processing station from the retrieval conveyance system, the selected input bin containing one or more objects; using a programmable motion device, grasping and removing a selected object of the one or more objects from the selected input bin; moving the selected object to a carrier using the programmable motion device; moving the carrier; providing an array of destination locations below the carrier such that the carrier is movable at an elevation above the array of destination locations; and dropping the selected object into a selected destination location of the array of destination locations by tipping the carrier to discharge the selected object into the selected destination location.
22. The method as claimed in claim 21, wherein the programmable motion device includes an articulated arm with a vacuum end-effector.
23. The method as claimed in claim 21, wherein the array of destination locations is provided as at least one pair of rows of destination locations.

24. The method as claimed in claim 21, wherein each destination location is provided adjacent an output conveyance system for receiving completed destination containers.

25. The method as claimed in claim 24, wherein the method further includes displacing the completed destination containers onto the output conveyance system using an output bin displacement system.

26. The method as claimed in claim 21, wherein the method further includes retrieving the plurality of input bins from the input bin storage section on the retrieval conveyance system.

27. The method as claimed in claim 26, wherein the retrieval conveyance system includes an infeed conveyor section that brings any of the plurality of input bins to the processing station.

28. The method as claimed in claim 27, wherein the retrieval conveyance system includes a bin displacement system for displacing the selected input bin onto the infeed conveyor section.

29. The method as claimed in claim 28, wherein the retrieval conveyance system includes a return conveyor section that returns the selected input bin to the input bin storage section.
